

Are Choice Experiments Incentive Compatible?

A Replication of Lusk & Schroeder (2004) in R

Jun-Jie (Justin) Chang · B12607021

2026-06-12

Table of contents

Preface	4
What is reproduced	4
How to read it	4
1 The original study	6
1.1 Research question	6
1.2 Empirical design	6
1.3 Key findings of the paper	7
2 Data structure and preparation	8
2.1 The raw file	8
2.2 The price design (paper appendix)	8
2.3 Cleaning: from spreadsheet to respondents	9
2.4 Reshape: wide → long → choice format	10
3 Descriptive replication: Tables 1 and 2	12
3.1 Table 1 — demographics by treatment	12
3.2 Table 2 — choice frequencies and the “None” gap	13
4 Multinomial logit and the scale test	15
4.1 Specification	15
4.2 Estimation code (R/03_table3_mnl.R)	15
4.3 Results	16
5 Testing IIA with the universal logit	18
5.1 Specification	18
5.2 Estimation code (R/06_iaa_universal_logit.R)	18
5.3 Results	18
6 Advanced models: HEV and RPL	20
6.1 Specifications	20
6.2 Estimation code (R/04_table4_adv.R)	20
6.3 Results	21
7 Willingness to pay and market shares	23
7.1 Specification	23
7.2 Estimation code (R/05_table5_wtp.R)	23
7.3 Table 5 — WTP and the Poe test	23
7.4 Figure 2 — predicted market shares (HEV)	24

8	Replication experience and discrepancies	26
8.1	Locating data and documentation	26
8.2	Translating LIMDEP $\rightarrow$ R	26
8.3	Interpreting the discrepancies	26
9	Reproducibility	28
9.1	How to rebuild everything	28
9.2	Session information	28
	References	29

Preface

This site is my replication, in **R**, of

Lusk, J. L., & Schroeder, T. C. (2004). *Are Choice Experiments Incentive Compatible? A Test with Quality Differentiated Beef Steaks*. American Journal of Agricultural Economics, **86**(2), 467–482.

The authors ran their analysis in LIMDEP. I redid it in R for the AGECE 5045 (Applied Discrete Choice Analysis, NTU 114-2) replication project. I wrote it as a Quarto book so that the code, the output it produces, and my own comments are all in one place and easy to follow.

What is reproduced

Output	Chapter	Method
Data preparation	Data	read <code>harrison.xls</code> , reshape to choice format
Tables 1–2	Descriptives	demographics & choice frequencies
Table 3	MNL	multinomial logit + Swait–Louviere scale test
IIA test	IIA	universal (“mother”) logit LR test
Table 4	Advanced models	HEV, RPL (MNP discussed)
Table 5 & Figure 2	WTP	willingness to pay + Poe test; HEV market shares
Discussion	Discrepancies	what went differently, and why

How to read it

The [Data](#) chapter runs for real when the book is built: it opens the raw spreadsheet and builds the estimation dataset step by step.

The model chapters work a little differently. The heavy models (the custom-likelihood MNL scale test, the HEV quadrature, the simulated-ML RPL, and the Poe bootstrap) are estimated by the numbered scripts in R/, which save their results to `output/*.rds`. The book shows that estimation code but loads the saved results rather than re-running them, so it builds in a few seconds. The lighter summaries and Figure 2 are still computed on the spot.

My own comments are in coloured callout boxes, so they are easy to tell apart from the paper's claims and from the raw output.

i Reproducibility

All code and data are in the [GitHub repository](#). To rebuild from scratch, run the scripts in R/ in order (see [Reproducibility](#)) and re-render the book.

If there is one result to remember, it is this: hypothetical choices overstate what people will really pay. Every qualitative finding of the paper comes through in my replication. The numbers that do not match exactly are listed and discussed in the final chapter.

1 The original study

1.1 Research question

In a choice experiment, people pick repeatedly among products that differ in their attributes and prices. The method is now standard for valuing food quality, environmental goods, and new products, and it is often assumed to be more incentive compatible than open-ended contingent valuation because it looks more like ordinary shopping.

Lusk and Schroeder ask whether that assumption actually holds. Do the choices people make, and the willingness-to-pay (WTP) numbers we back out of them, change when the task is hypothetical rather than backed by real money? This is the “hypothetical bias” question from the contingent valuation literature, asked inside the choice-experiment setting. What makes the paper useful is that it does not just ask whether bias exists; it tries to say which WTP measure is affected, the marginal value of a quality change or the total value of a steak versus buying nothing.

1.2 Empirical design

The good is a 12-oz ribeye steak, offered under five quality or assurance labels: Generic, Guaranteed Tender (GT), Natural, USDA Choice, and Certified Angus Beef (CAB). A sixth option, “None”, lets the respondent decline to buy, so each choice is among six alternatives.

Each subject saw 17 scenarios. In every scenario the five steaks were offered at prices that varied by design (from \$3.38 to \$9.00), and the subject picked one steak or chose None. The full price design is in the paper’s appendix and is reproduced in the [Data](#) chapter.

The manipulation that matters is the treatment:

- In the *nonhypothetical* (real) treatment, subjects knew that one scenario would be drawn at random and would actually count: they would buy the steak they had chosen at the posted price. Scenario 17, where all five steaks cost the same \$5.63, was the binding scenario.
- The *hypothetical* treatment used the identical task, except no purchase ever happened.

For estimation the authors fit a series of discrete-choice models to the panel of choices: a multinomial logit (MNL), a test of IIA based on the universal (“mother”) logit, and three less restrictive models, namely heteroscedastic extreme value (HEV), multinomial probit (MNP), and random parameters logit (RPL). They compare the two treatments with the Swait–Louviere scale-and-preference test and test WTP differences with the Poe combinatorial test. All of this was done in LIMDEP.

1.3 Key findings of the paper

The paper reports four main results. Hypothetical respondents choose the no-purchase option less often and have higher WTP than the real-payment group, so hypothetical bias is present. The error scale differs across treatments, and a joint model rejects equality of preferences across the two groups. IIA is violated for the pooled and hypothetical data, which is what motivates the richer HEV, MNP, and RPL models. Finally, the bias shows up mainly in total WTP (a steak versus buying nothing) rather than in the marginal premiums between quality levels, which is the practically useful part of the result.

i Why this paper is good to replicate

Almost every headline number is a likelihood, a test statistic, or a WTP, so each one can be recomputed from the raw choices. That means the hard part is rarely deciding what to calculate. The hard part is matching LIMDEP's 2004 defaults in present-day R, and that is where most of my discrepancies come from (see [Discrepancies](#)).

2 Data structure and preparation

This chapter actually runs when the book is built. It reads the raw spreadsheet and builds the estimation dataset, following the same data-preparation steps we used in class.

2.1 The raw file

The data came as `harrison.xls`. The sheet I need, `trt 10&11`, is in wide form: one row per respondent, with a two-row header, label rows that separate the two treatments, and a few summary rows at the bottom. Treatment 10 is the nonhypothetical group and Treatment 11 is the hypothetical group.

```
raw_dims <- dim(readxl::read_excel("harrison.xls", sheet = "trt 10&11",
                                   skip = 3, col_types = "text"))
cat("Raw sheet (after skipping 3 header rows):",
    raw_dims[1], "rows x", raw_dims[2], "columns\n")
#> Raw sheet (after skipping 3 header rows): 129 rows x 57 columns
```

The columns fall into a few groups:

Block	Columns	Contents
Identifiers	subject, session	Subject ID (1001–1099 = nonhyp; 1100–1199 = hyp)
Demographics	gender, age, education, student, income, ...	Used for Table 1
Attitudes	color, tenderness, flavor, ...	Survey Likert items (not used in estimation)
Choices	sc1–sc17	Chosen alternative per scenario, coded 1–6

The choice codes are 1 = Generic (GEN), 2 = Guaranteed Tender (GT), 3 = Natural (NAT), 4 = USDA Choice (CHO), 5 = CAB, 6 = None (NONE).

2.2 The price design (paper appendix)

The 17×5 price matrix is the experimental design. It is the same for both treatments, and it is what connects a stored choice *code* to the *prices* the respondent was actually looking at.


```
price_matrix <- tibble::tribble(
  ~scenario, ~GEN, ~GT, ~NAT, ~CHO, ~CAB,
  1, 3.38, 7.88, 6.75, 6.75, 6.75, 2, 4.50, 9.00, 6.75, 5.63, 9.00,
  3, 6.75, 9.00, 5.63, 9.00, 6.75, 4, 5.63, 9.00, 9.00, 6.75, 5.63,
  5, 6.75, 6.75, 6.75, 7.88, 7.88, 6, 4.50, 7.88, 7.88, 9.00, 9.00,
  7, 4.50, 6.75, 5.63, 6.75, 7.88, 8, 5.63, 5.63, 5.63, 7.88, 9.00,
  9, 5.63, 7.88, 5.63, 7.88, 9.00, 10, 5.63, 6.75, 7.88, 5.63, 9.00,
  11, 6.75, 7.88, 9.00, 5.63, 7.88, 12, 4.50, 5.63, 5.63, 7.88, 7.88,
  13, 3.38, 9.00, 7.88, 7.88, 9.00, 14, 3.38, 5.63, 5.63, 5.63, 5.63,
  15, 3.38, 6.75, 9.00, 9.00, 9.00, 16, 6.75, 5.63, 7.88, 6.75, 6.75,
  17, 5.63, 5.63, 5.63, 5.63, 5.63
)
kable(price_matrix, caption = "Price design: 17 scenarios × 5 steaks (USD).")
```

Table 2.2: Price design: 17 scenarios × 5 steaks (USD).

scenario	GEN	GT	NAT	CHO	CAB
1	3.38	7.88	6.75	6.75	6.75
2	4.50	9.00	6.75	5.63	9.00
3	6.75	9.00	5.63	9.00	6.75
4	5.63	9.00	9.00	6.75	5.63
5	6.75	6.75	6.75	7.88	7.88
6	4.50	7.88	7.88	9.00	9.00
7	4.50	6.75	5.63	6.75	7.88
8	5.63	5.63	5.63	7.88	9.00
9	5.63	7.88	5.63	7.88	9.00
10	5.63	6.75	7.88	5.63	9.00
11	6.75	7.88	9.00	5.63	7.88
12	4.50	5.63	5.63	7.88	7.88
13	3.38	9.00	7.88	7.88	9.00
14	3.38	5.63	5.63	5.63	5.63
15	3.38	6.75	9.00	9.00	9.00
16	6.75	5.63	7.88	6.75	6.75
17	5.63	5.63	5.63	5.63	5.63

2.3 Cleaning: from spreadsheet to respondents

I keep only the rows whose subject ID is between 1001 and 1199. That filter alone drops the treatment-label rows and the summary rows at the bottom. One more case needs handling: subject 1060 has no answers for any of the 17 scenarios, so I drop it. Doing this brings the file's 68 rows down to the 67 nonhypothetical subjects the paper reports.

```
demo_names <- c(
  "subject", "gender", "age", "household", "child", "education", "income", "student",
```

```

"occupation", "race", "gbeef", "steak_buy", "other_beef", "gbeef_freq", "steak_freq",
"color", "brand", "quality_gd", "ex_fat", "in_fat", "price_imp", "safety", "juiciness",
"flavor", "tenderness", "consistency", "doneness", "production", "quality_gd2",
"slaughter", "food_safe", "quality_pref", "cooked", "chance1", "chance2", "chance3",
"chance4", "chance5")
sc_names <- paste0("sc", 1:17)

raw <- read_excel("harrison.xls", sheet = "trt 10&11",
  col_names = c(demo_names, sc_names, "blank", "session"),
  skip = 3, col_types = "text")

df_wide <- raw |>
  mutate(subject = suppressWarnings(as.numeric(subject))) |>
  filter(!is.na(subject), subject >= 1001, subject <= 1199) |>
  filter(subject != 1060) |> # all scenarios missing
  mutate(treatment = factor(if_else(subject < 1100, "nonhyp", "hyp"),
    levels = c("nonhyp", "hyp"))) |>
  mutate(across(all_of(c(setdiff(demo_names, "subject"), sc_names, "session")),
    as.numeric))

cat("Nonhypothetical subjects:", sum(df_wide$treatment == "nonhyp"), "(paper: 67)\n")
#> Nonhypothetical subjects: 67 (paper: 67)
cat("Hypothetical subjects:", sum(df_wide$treatment == "hyp"), "(paper: 37)\n")
#> Hypothetical subjects: 37 (paper: 37)

```

2.4 Reshape: wide → long → choice format

Two reshapes turn the respondent table into something the choice models can use. First I go to long form, one row per respondent and scenario, and merge in the five prices. Then I expand to choice (mlogit) form, one row per respondent, scenario, and alternative, with a logical chosen flag and the alternative's own price. Scenario 17 is dropped at this stage, since it is the all-equal-price binding scenario, which leaves scenarios 1 to 16 for estimation.

```

df_long <- df_wide |>
  select(subject, treatment, session, all_of(sc_names)) |>
  pivot_longer(all_of(sc_names), names_to = "sc_var", values_to = "choice_code") |>
  mutate(scenario = as.integer(sub("sc", "", sc_var)),
    choice = factor(choice_code, levels = 1:6, labels = alt_labels)) |>
  select(-sc_var, -choice_code) |>
  left_join(price_matrix, by = "scenario") |>
  arrange(subject, scenario)

df_mlogit <- df_long |>
  filter(scenario <= 16) |> # drop binding scenario
  crossing(alt = factor(alt_labels, levels = alt_labels)) |>
  mutate(

```

```

chosen    = as.character(choice) == as.character(alt),
price_alt = case_when(
  alt == "GEN" ~ GEN, alt == "GT" ~ GT, alt == "NAT" ~ NAT,
  alt == "CHO" ~ CHO, alt == "CAB" ~ CAB, alt == "NONE" ~ NA_real_),
chid = paste(subject, scenario, sep = "_") |>
select(chid, subject, treatment, session, scenario, alt, chosen, price_alt) |>
arrange(subject, scenario, alt)

cat("Nonhyp choice situations:",
  sum(df_mlogit$treatment == "nonhyp" & df_mlogit$alt == "GEN"), "(paper: 1072 = 67×16)\n")
#> Nonhyp choice situations: 1072 (paper: 1072 = 67×16)
cat("Hyp choice situations:",
  sum(df_mlogit$treatment == "hyp" & df_mlogit$alt == "GEN"), "(paper: 592 = 37×16)\n")
#> Hyp choice situations: 592 (paper: 592 = 37×16)

df_mlogit |> filter(chid == "1001_1") |>
select(chid, alt, chosen, price_alt) |>
kable(caption = "One choice situation in long format (subject 1001, scenario 1).")

```

Table 2.3: One choice situation in long format (subject 1001, scenario 1).

chid	alt	chosen	price_alt
1001_1	GEN	FALSE	3.38
1001_1	GT	FALSE	7.88
1001_1	NAT	FALSE	6.75
1001_1	CHO	FALSE	6.75
1001_1	CAB	FALSE	6.75
1001_1	NONE	TRUE	NA

i Two things I learned here

The “68 vs 67” mismatch is the first thing you run into. Instead of guessing which subject to drop, I looked for a reason and found that subject 1060 is empty, which is the kind of note the original LIMDEP file simply does not carry.

The other point is that **NONE** has no price. Coding its `price_alt` as **NA** or 0, and never giving it an own-price coefficient, is what makes it the natural base alternative.

3 Descriptive replication: Tables 1 and 2

3.1 Table 1 — demographics by treatment

Table 1 in the paper is a randomisation check: it shows that the two treatment groups look alike on observables. I reproduce the means, standard deviations, and the t-tests for equality of means.

```
mean_sd <- function(data, var) {
  data |> group_by(treatment) |>
    summarise(mean = round(mean(.data[[var]], na.rm = TRUE), 2),
              sd   = round(sd(.data[[var]], na.rm = TRUE), 2), .groups = "drop") |>
    pivot_wider(names_from = treatment, values_from = c(mean, sd)) |>
    mutate(variable = var)
}
t1_vars <- c("gender", "age", "education", "student", "income")
table1 <- bind_rows(lapply(t1_vars, mean_sd, data = df_wide)) |>
  rowwise() |>
  mutate(t_p = round(t.test(df_wide[[variable]][df_wide$treatment == "nonhyp"],
                           df_wide[[variable]][df_wide$treatment == "hyp"])$p.value, 3)) |>
  ungroup() |>
  select(variable, mean_nonhyp, sd_nonhyp, mean_hyp, sd_hyp, t_p)

kable(table1,
      col.names = c("Variable", "NH mean", "NH (SD)", "HY mean", "HY (SD)", "t-test p"),
      caption = "Table 1. Demographics by treatment (NH = nonhypothetical, HY = hypothetical).")
```

Table 3.1: Table 1. Demographics by treatment (NH = nonhypothetical, HY = hypothetical).

Variable	NH mean	NH (SD)	HY mean	HY (SD)	t-test p
gender	0.42	0.50	0.41	0.50	0.903
age	33.13	17.55	29.62	14.92	0.284
education	3.66	2.06	3.38	1.98	0.500
student	0.51	0.50	0.59	0.50	0.397
income	4.63	4.82	3.76	4.06	0.332

Result

None of the t-tests is significant at 5%, so I cannot reject equality of the demographic means across treatments, just as in the paper. Because the groups are balanced, any difference

in behaviour later on can be read as coming from the hypothetical-versus-real manipulation rather than from who happened to be in each group.

3.2 Table 2 — choice frequencies and the “None” gap

```
table2 <- df_long |>
  count(treatment, scenario, choice) |>
  group_by(treatment, scenario) |>
  mutate(pct = round(100 * n / sum(n), 1)) |>
  ungroup() |> select(-n) |>
  pivot_wider(names_from = choice, values_from = pct, values_fill = 0) |>
  select(treatment, scenario, all_of(alt_labels)) |>
  arrange(treatment, scenario)

kable(table2 |> filter(treatment == "nonhyp") |> select(-treatment),
      col.names = c("Scenario", "Generic", "GT", "Natural", "USDA Choice", "CAB", "None"),
      caption = "Table 2a. % choosing each alternative – Nonhypothetical.")
```

Table 3.2: Table 2a. % choosing each alternative — Nonhypothetical.

Scenario	Generic	GT	Natural	USDA Choice	CAB	None
1	28.4	1.5	4.5	19.4	20.9	25.4
2	10.4	1.5	1.5	65.7	1.5	19.4
3	0.0	0.0	29.9	0.0	32.8	37.3
4	0.0	1.5	0.0	7.5	74.6	16.4
5	0.0	3.0	3.0	1.5	76.1	16.4
6	10.4	1.5	1.5	1.5	68.7	16.4
7	16.4	9.0	16.4	25.4	0.0	32.8
8	0.0	61.2	6.0	0.0	3.0	29.9
9	11.9	3.0	25.4	16.4	1.5	41.8
10	0.0	0.0	1.5	73.1	7.5	17.9
11	0.0	1.5	0.0	79.1	1.5	17.9
12	11.9	38.8	0.0	1.5	17.9	29.9
13	40.3	0.0	1.5	10.4	14.9	32.8
14	10.4	13.4	6.0	13.4	44.8	11.9
15	34.3	31.3	0.0	0.0	3.0	31.3
16	0.0	50.7	3.0	16.4	0.0	29.9
17	0.0	7.5	6.0	19.4	52.2	14.9

```
kable(table2 |> filter(treatment == "hyp") |> select(-treatment),
      col.names = c("Scenario", "Generic", "GT", "Natural", "USDA Choice", "CAB", "None"),
      caption = "Table 2b. % choosing each alternative – Hypothetical.")
```

Table 3.3: Table 2b. % choosing each alternative — Hypothetical.

Scenario	Generic	GT	Natural	USDA Choice	CAB	None
1	48.6	0.0	10.8	13.5	24.3	2.7
2	10.8	0.0	2.7	86.5	0.0	0.0
3	0.0	2.7	48.6	0.0	37.8	10.8
4	2.7	0.0	0.0	5.4	83.8	8.1
5	0.0	13.5	10.8	0.0	67.6	8.1
6	21.6	0.0	0.0	0.0	75.7	2.7
7	24.3	13.5	40.5	21.6	0.0	0.0
8	0.0	64.9	16.2	2.7	2.7	13.5
9	16.2	5.4	51.4	13.5	0.0	13.5
10	2.7	0.0	0.0	89.2	0.0	8.1
11	0.0	0.0	0.0	91.9	0.0	8.1
12	13.5	59.5	2.7	0.0	16.2	8.1
13	56.8	0.0	0.0	5.4	21.6	16.2
14	32.4	10.8	13.5	21.6	21.6	0.0
15	48.6	32.4	0.0	0.0	5.4	13.5
16	0.0	59.5	0.0	27.0	0.0	13.5
17	0.0	18.9	13.5	32.4	32.4	2.7

```

none_gap <- table2 |> filter(scenario <= 16) |>
  select(treatment, scenario, NONE) |>
  pivot_wider(names_from = treatment, values_from = NONE) |>
  mutate(gap = nonhyp - hyp)
cat(sprintf("Smallest None gap: scenario %d (%.1f pp)\n",
  none_gap$scenario[which.min(abs(none_gap$gap))], min(abs(none_gap$gap))))
#> Smallest None gap: scenario 4 (8.3 pp)
cat(sprintf("Largest None gap: scenario %d (%.1f pp)\n",
  none_gap$scenario[which.max(none_gap$gap)], max(none_gap$gap)))
#> Largest None gap: scenario 7 (32.8 pp)
cat("Paper: smallest = scenario 4 (8.31 pp), largest = scenario 7 (32.84 pp)\n")
#> Paper: smallest = scenario 4 (8.31 pp), largest = scenario 7 (32.84 pp)

```

💡 Result: the basic pattern

In every scenario, the nonhypothetical group picks “None” more often than the hypothetical group. When the payment is real, more people simply walk away. The size of that gap lines up with the paper, which is largest at scenario 7 and smallest at scenario 4.

4 Multinomial logit and the scale test

How the model chapters work

From here on the estimation code is shown but not re-run when the book builds. It is the code in R/03–R/06. The fitted objects are loaded from `output/`, and the results are recomputed for display. Each chapter follows the same order: specification, estimation code, results, then my comments.

4.1 Specification

The utility of alternative j in choice situation i is

$$V_{ij} = \beta_j + \alpha_j P_{ij}, \quad j \in \{\text{GEN, GT, NAT, CHO, CAB}\}, \quad V_{i,\text{NONE}} \equiv 0.$$

So there are five alternative-specific constants β_j and five own-price coefficients α_j , ten taste parameters in all. The joint model stacks both treatments, forces the taste parameters to be equal, and scales the nonhypothetical utilities by μ :

$$V_{ij}^{\text{NH}} = \mu (\beta_j + \alpha_j P_{ij}), \quad V_{ij}^{\text{H}} = \beta_j + \alpha_j P_{ij}.$$

Here μ is the error scale of the real-payment group relative to the hypothetical group. The Swait–Louviere LR test compares this joint model with the two separate models.

4.2 Estimation code (R/03_table3_mnl.R)

```
# Alternative-specific price variables: p_j = price × 1{alt = j}
df0 <- df_mlogit |>
  mutate(price_alt = tidyr::replace_na(price_alt, 0),
         p_GEN = if_else(alt=="GEN", price_alt, 0),
         p_GT  = if_else(alt=="GT",  price_alt, 0),
         p_NAT = if_else(alt=="NAT", price_alt, 0),
         p_CHO = if_else(alt=="CHO", price_alt, 0),
         p_CAB = if_else(alt=="CAB", price_alt, 0))

make_mdata <- function(d) mlogit.data(d, choice="chosen", shape="long",
```

```

alt.var="alt", chid.var="chid", id.var="subject", alt.levels=alts)

f_mnl <- chosen ~ p_GEN + p_GT + p_NAT + p_CHO + p_CAB | 1 # 5 prices + 5 ASCs
mnl_nh <- mlogit(f_mnl, make_mdata(filter(df0, treatment=="nonhyp")), reflevel="NONE")
mnl_hyp <- mlogit(f_mnl, make_mdata(filter(df0, treatment=="hyp")), reflevel="NONE")

# Joint model with scale mu: custom log-likelihood maximised by optim(BFGS).
# theta = (beta[1:5], alpha[6:10], log_mu[11]); V_NH scaled by exp(log_mu).
ll_joint_fn <- function(theta, df) {
  beta <- c(theta[1:5], 0); alpha <- c(theta[6:10], 0); mu <- exp(theta[11])
  k <- match(as.character(df$alt), alts)
  V <- beta[k] + alpha[k]*df$price_alt
  V[df$treatment=="nonhyp"] <- mu * V[df$treatment=="nonhyp"]
  df |> mutate(V=V) |> group_by(chid) |>
    mutate(p = exp(V-max(V))/sum(exp(V-max(V)))) |> ungroup() |>
    summarise(ll = sum(log(pmax(p[chosen], 1e-300)))) |> pull(ll)
}
opt <- optim(theta0, ll_joint_fn, df=df0, method="BFGS",
  control=list(fnscale=-1, maxit=1e4, reltol=1e-12), hessian=TRUE)

# Swait-Louviere LR scale test: chi2 = -2(LL_joint - LL_nh - LL_hyp)
lr_stat <- -2 * (opt$value - logLik(mnl_nh) - logLik(mnl_hyp))

```

There is no off-the-shelf `mlogit` command for the scaled joint model, so I wrote the log-likelihood by hand and maximised it with `optim`. The $\exp(V - \max(V))$ inside the probability is just the log-sum-exp trick to keep things numerically stable.

4.3 Results

```

kable(t3$table3 |> mutate(Parameter = gk(Parameter)),
  col.names = c("Parameter", "NH Est", "NH (SE)", "HY Est", "HY (SE)", "JT Est", "JT (SE)"),
  caption = "Table 3. MNL estimates – Nonhyp | Hyp | Joint (with scale).")

```

Table 4.1: Table 3. MNL estimates — Nonhyp | Hyp | Joint (with scale).

Parameter	NH Est	NH (SE)	HY Est	HY (SE)	JT Est	JT (SE)
beta GEN (Generic)	3.9094	0.5227	5.8868	0.6536	5.2285	0.5137
beta GT (Guaranteed Tender)	10.0676	0.8867	12.5954	1.1989	12.4289	0.9639
beta NAT (Natural)	5.9844	1.1660	11.1365	1.6456	9.3556	1.2035
beta CHO (USDA Choice)	9.2681	0.6670	13.4089	1.0984	12.1501	0.8534
beta CAB (Certified Angus Beef)	4.5184	0.4932	5.1333	0.7168	5.2656	0.5034
alpha GEN (price)	-1.0880	0.1284	-1.1654	0.1541	-1.2759	0.1224
alpha GT (price)	-1.6216	0.1435	-1.7957	0.1916	-1.9218	0.1522

Parameter	NH Est	NH (SE)	HY Est	HY (SE)	JT Est	JT (SE)
alpha NAT (price)	-1.1629	0.1949	-1.7176	0.2798	-1.6250	0.2023
alpha CHO (price)	-1.3998	0.1032	-1.8344	0.1699	-1.7724	0.1278
alpha CAB (price)	-0.5974	0.0641	-0.5294	0.0910	-0.6490	0.0637

```

mnl_summary <- tibble(
  Statistic = c("Log-likelihood", "Observations", "Scale (mu)", "SE(mu)",
    "Scale-test Chi-sq", "df", "p-value"),
  Nonhyp = c(fmt(t3$ll_nh,1), "1072", "-", "-", "-", "-", "-"),
  Hyp = c(fmt(t3$ll_hyp,1), "592", "-", "-", "-", "-", "-"),
  Joint = c(fmt(t3$ll_joint,1), "1664", fmt(t3$mu_hat,3), fmt(t3$se_mu,3),
    fmt(t3$lr_stat,1), as.character(t3$df_test), formatC(t3$p_lr, format="e", digits=1)))
kable(mnl_summary, caption = "Table 3 fit statistics.")

```

Table 4.2: Table 3 fit statistics.

Statistic	Nonhyp	Hyp	Joint
Log-likelihood	-1400.3	-677.9	-2131.2
Observations	1072	592	1664
Scale (mu)	—	—	0.804
SE(mu)	—	—	0.056
Scale-test Chi-sq	—	—	105.9
df	—	—	9
p-value	—	—	9.9e-19

⚠ Where this differs from the paper

My scale estimate is $\hat{\mu} = 0.804$ (SE 0.056), against the paper's 0.90 (SE 0.06). Both are far enough from 1 to be significant, and both say the two treatments have different error scales, but the point estimates are not the same. My scale test is $\chi^2 = 105.9$ on 9 df, while the paper reports 106.8 on 4 df. The χ^2 values are close. The degrees of freedom differ because of how the joint restriction is counted: I restrict 10 taste parameters and free 1 scale, which gives $(10 + 10) - 11 = 9$, whereas the paper's "4" implies a different count. Either way the conclusion is the same, the pooling is rejected.

One more thing is worth flagging. My log-likelihoods are a bit lower than the paper's (nonhyp -1400.3 against -1261.4 ; hyp -677.9 against -627.4). The same offset shows up again in the [IIA test](#), and it is the one loose end I could not fully tie off. My best guess is that the price enters utility a little differently than in the original; I come back to this in [Discrepancies](#).

5 Testing IIA with the universal logit

5.1 Specification

The universal, or “mother”, logit allows the cross-price effects that MNL rules out:

$$V_{ij} = \beta_j + \sum_{k=1}^5 \alpha_{jk} P_{ik}.$$

That is 30 parameters: 5 constants plus a full 5×5 price matrix. MNL is the special case where $\alpha_{jk} = 0$ for $j \neq k$, which is 20 restrictions, so the IIA likelihood-ratio test has 20 degrees of freedom. A second test takes the relaxed model and asks whether preferences are equal across treatments, comparing the joint universal logit with the two separate ones (30 degrees of freedom).

5.2 Estimation code (R/06_iaa_universal_logit.R)

```
# 25 cross-price dummies: p_{j}_{k} = P_k × 1{alt = j}; zero for NONE.
for (j in steak_alts) for (k in steak_alts)
  df0[[paste0("p_",j,"_",k)]] <- ifelse(df0$alt==j, df0[[paste0("P_",k)]], 0)

f_ul <- as.formula(paste("chosen ~", paste(ul_price_names, collapse=" + "), "| 1"))
ul_nh <- mlogit(f_ul, make_mdata(filter(df0, treatment=="nonhyp")), reflevel="NONE")
ul_hyp <- mlogit(f_ul, make_mdata(filter(df0, treatment=="hyp")), reflevel="NONE")
# Joint UL with scale via custom optim (31 params); see script.

chi2 <- function(ll_ul, ll_mnl) 2 * (ll_ul - ll_mnl) # IIA LR, df = 20
```

5.3 Results

```
kable(iaa$summary_tbl,
      col.names = c("Model", "LL(MNL)", "LL(UL)", "Chi-sq(20)", "p-value", "Paper Chi-sq", "Paper p"),
      caption = "IIA test via universal logit (cf. Lusk & Schroeder 2004, p. 476).")
```

Table 5.1: IIA test via universal logit (cf. Lusk & Schroeder 2004, p. 476).

Model	LL(MNL)	LL(UL)	Chi-sq(20)	p-value	Paper Chi-sq	Paper p
Joint	-2131.2	-2048.1	166.2	0	66.9	<0.01
Nonhypothetical	-1400.3	-1349.8	101.1	0	31.3	0.051
Hypothetical	-677.9	-634.8	86.2	0	59.5	<0.01

```
cat(sprintf("Preference-equality test (relaxed model): Chi-sq(%d) = %.1f, p = %.4f [paper: Chi-sq(30)=127.0, p=0.0000]"),
    iia$df_eq, iia$chi2_eq, iia$p_eq)
#> Preference-equality test (relaxed model): Chi-sq(30) = 127.0, p = 0.0000 [paper: Chi-sq(30)=130.7]
```

⚠ Where this differs from the paper

The conclusion is the same as the paper: IIA is rejected in all three samples. But my test statistics are larger. My $\chi^2(20)$ values are 166 (joint), 101 (nonhyp), and 86 (hyp), against the paper's 66.9, 31.3, and 59.5. The nonhyp case matters most here, because the paper's 31.3 is only borderline ($p \approx 0.051$) while mine is clearly significant.

The reason is in the log-likelihoods. My LL levels are consistently lower than the paper's (for example nonhyp MNL -1400.3 against -1261.4, and joint UL -2048.1 against -1908.8). A roughly constant gap in the LL level pushes up the IIA χ^2 , which is a difference of two of these offset likelihoods within one sample. It should not affect a test that differences across samples, and indeed it does not: my preference-equality test comes out at $\chi^2(30) = 127.0$, almost exactly the paper's 130.7. So the problem is in the base MNL/UL specification, most likely the way price enters utility, rather than in the IIA test itself. More on this in [Discrepancies](#).

i Why IIA matters here

IIA says the relative odds of any two alternatives do not depend on the others. That is the price of using MNL. Rejecting it, especially in the hypothetical and pooled samples, is what justifies the richer models in the next chapter: people are not treating the steaks as independent draws, so a model that lets the error scales differ across alternatives fits better.

6 Advanced models: HEV and RPL

6.1 Specifications

Three models go into Table 4, and I fit two of them.

HEV (heteroscedastic extreme value) is like MNL except that each alternative has its own Gumbel scale σ_j , with $\sigma_{\text{NONE}} = 1$ fixed for identification. Letting the error variances differ by alternative is how it relaxes IIA. I fit it with a hand-written log-likelihood that integrates the choice probability over the Gumbel density using trapezoidal quadrature.

RPL (random parameters logit) lets the constants vary across respondents as normal draws (a mean and a standard deviation each), while the price coefficients stay fixed so the model converges. I estimate it by simulated maximum likelihood with 500 draws in `mlogit`.

MNP (multinomial probit) sets the off-diagonal correlations to zero and fixes two standard deviations to 1. I do not fit it here. It needs the `apollo` package and a GHK simulator over a five-dimensional normal, which is more than this project took on, so I describe the intended setup rather than report estimates.

6.2 Estimation code (R/04_table4_adv.R)

```
## RPL: ASCs random normal, prices fixed; simulated ML with 500 draws
rpar_spec <- setNames(rep("n", 5), asc_names)
rpl_nh <- mlogit(f_mnl, mdata_nh, refllevel="NONE", rpar=rpar_spec, R=500, halton=NA, panel=TRUE)
rpl_hyp <- mlogit(f_mnl, mdata_hyp, refllevel="NONE", rpar=rpar_spec, R=500, halton=NA, panel=TRUE)

## HEV: per-alternative scale sigma_j (sigma_NONE = 1). The probability that j is
## chosen is the integral over the Gumbel density of alternative j of the product
## of the other alternatives' CDFs; evaluated on a trapezoidal grid w in [-8, 45]
## and maximised by optim(BFGS).
opt_hev_nh <- optim(theta0_nh, hev_ll, df_s=df_nh, method="BFGS",
                    control=list(fnscale=-1, reltol=1e-10), hessian=TRUE)
opt_hev_hyp <- optim(theta0_hyp, hev_ll, df_s=df_hyp, method="BFGS",
                    control=list(fnscale=-1, reltol=1e-10), hessian=TRUE)
```

6.3 Results

```
rpl_fit <- tibble(
  Model = c("RPL Nonhypothetical (R=500)", "RPL Hypothetical (R=500)"),
  `Log-likelihood` = c(fmt(t4r$ll_nh, 1), fmt(t4r$ll_hyp, 1)))
kable(rpl_fit, caption = "Table 4a. Random parameters logit fit.")
```

Table 6.1: Table 4a. Random parameters logit fit.

Model	Log-likelihood
RPL Nonhypothetical (R=500)	-1053.6
RPL Hypothetical (R=500)	-571.3

```
kable(t4h$table4_hev |> mutate(Parameter = gk(Parameter)),
  col.names = c("Parameter", "NH Est", "NH (SE)", "HY Est", "HY (SE)", "JT Est", "JT (SE)"),
  caption = "Table 4b. HEV estimates – taste (beta, alpha) and scale (sigma) parameters. sigma(NONE) = 1 fixed.")
```

Table 6.2: Table 4b. HEV estimates — taste (beta, alpha) and scale (sigma) parameters. sigma(NONE) = 1 fixed.

Parameter	NH Est	NH (SE)	HY Est	HY (SE)	JT Est	JT (SE)
beta GEN (ASC)	2.8329	0.1442	5.6871	0.1245	4.1902	0.0000
beta GT (ASC)	3.4054	0.0000	5.8381	0.2129	4.7900	0.0000
beta NAT (ASC)	3.3112	0.0000	6.9589	0.0000	5.1367	0.0000
beta CHO (ASC)	2.8377	0.0143	4.8206	0.0372	3.9721	0.0000
beta CAB (ASC)	2.1223	0.0000	3.5628	0.0465	2.9722	0.0090
alpha GEN (price)	-0.6699	0.0473	-1.1479	0.0022	-0.9011	0.0000
alpha GT (price)	-0.4241	0.0000	-0.6418	0.0371	-0.5761	0.0000
alpha NAT (price)	-0.4994	0.0000	-0.9184	0.0000	-0.7266	0.0000
alpha CHO (price)	-0.2727	0.0023	-0.3884	0.0032	-0.3629	0.0000
alpha CAB (price)	-0.1731	0.0000	-0.2119	0.0052	-0.2248	0.0000
sigma GEN (scale)	1.7589	0.1165	0.9309	0.0000	1.3182	0.0000
sigma GT (scale)	4.0009	0.0000	2.5415	0.1948	3.1729	0.0709
sigma NAT (scale)	2.5798	0.0962	1.8185	0.0552	2.1816	0.0439
sigma CHO (scale)	160.9314	88.7059	140.8183	90.5479	163.1633	68.7494
sigma CAB (scale)	4.9296	0.0000	6.7284	0.2972	4.6817	0.0617

```
cat(sprintf("HEV log-likelihood – NH: %.1f | HY: %.1f | Joint: %.1f\n",
  t4h$ll_nh, t4h$ll_hyp, t4h$ll_joint))
#> HEV log-likelihood – NH: -1345.0 | HY: -638.6 | Joint: -2047.1
```

 A convergence problem with HEV

The HEV scale for USDA Choice comes out far too large ($\hat{\sigma}_{\text{CHO}} \approx 161$ in the nonhyp model). That is a sign the variance of this alternative is only weakly identified given the price design. It feeds through to an inflated HEV WTP for Choice in the [next chapter](#), and it leaves the HEV covariance matrix close to singular, which is why I run it through `Matrix::nearPD` before the Poe bootstrap. Better starting values or tighter bounds (I already estimate $\log \sigma$ rather than σ) would be the next thing to try. This is the shakiest number in the whole replication.

7 Willingness to pay and market shares

7.1 Specification

Total WTP for steak j , relative to buying nothing, is

$$\text{WTP}_j = -\frac{\beta_j}{\alpha_j}.$$

To compare treatments I use the Poe combinatorial test, which asks whether nonhypothetical WTP is lower than hypothetical WTP (the hypothetical-bias hypothesis). For each treatment I draw 1000 coefficient vectors from $\mathcal{N}(\hat{\theta}, \widehat{\text{Var}})$, form all 1000×1000 differences in WTP, and take the one-tailed p-value as the share of those differences that fall below zero.

7.2 Estimation code (R/05_table5_wtp.R)

```
compute_wtp <- function(theta) -theta[1:5] / theta[6:10] # -beta_j / alpha_j

poe_test <- function(vcov_nh, vcov_hyp, theta_nh, theta_hyp, B = 1000) {
  draws_nh <- MASS::mvrnorm(B, theta_nh, make_pd(vcov_nh)) # PD repair if needed
  draws_hyp <- MASS::mvrnorm(B, theta_hyp, make_pd(vcov_hyp))
  wtp_nh <- t(apply(draws_nh, 1, compute_wtp))
  wtp_hyp <- t(apply(draws_hyp, 1, compute_wtp))
  sapply(1:5, function(j) # P(WTP_NH < WTP_HY)
    mean(sapply(wtp_nh[,j], function(w) mean(w < wtp_hyp[,j]))))
}
```

7.3 Table 5 — WTP and the Poe test

```
kable(t5$table5,
      col.names = c("Steak", "MNL NH", "MNL HY", "MNL p",
                    "HEV NH", "HEV HY", "HEV p", "RPL NH", "RPL HY", "RPL p"),
      caption = "Table 5. WTP vs. None (USD) and Poe one-tailed p-values, by model.")
```

Table 7.1: Table 5. WTP vs. None (USD) and Poe one-tailed p-values, by model.

Steak	MNL NH	MNL HY	MNL p	HEV NH	HEV HY	HEV p	RPL NH	RPL HY	RPL p
Generic	3.59	5.05	1	4.23	4.95	1	2.48	4.23	1.0000
Guaranteed Tender	6.21	7.01	1	8.03	9.10	1	5.80	6.67	1.0000
Natural	5.15	6.48	1	6.63	7.58	1	5.04	5.97	0.9999
USDA Choice	6.62	7.31	1	10.41	12.41	1	6.69	7.17	0.9982
Certified Angus Beef	7.56	9.70	1	12.26	16.81	1	7.71	9.18	1.0000

💡 Result: the main finding

For all three models and all five steaks, hypothetical WTP is higher than nonhypothetical WTP, and the Poe p-values sit at about 1, meaning it is essentially certain that $WTP_{NH} < WTP_{HY}$. That is the paper's central result: stated WTP overstates real WTP. The HEV numbers for Choice are inflated by the σ_{CHO} problem from the [previous chapter](#), so read those with that caveat.

7.4 Figure 2 — predicted market shares (HEV)

I rebuild Figure 2 here from the saved HEV parameters. For each scenario I work out the HEV choice probabilities and plot them across scenarios 1 to 16 for both treatments.

```
n_q <- 200; w_grid <- seq(-8, 45, length.out = n_q)
wt_trap <- c(0.5, rep(1, n_q - 2), 0.5) * diff(w_grid)[1]; ev_base <- exp(-w_grid)

hev_probs_single <- function(V, sigma) {
  J <- length(V); lp <- numeric(J)
  for (j in seq_len(J)) {
    oth <- seq_len(J)[-j]; r <- sigma[oth] / sigma[j]; d <- V[j] - V[oth]
    inner <- as.vector(exp(-sigma[oth] * d) %*% exp(outer(-r, w_grid)))
    li <- -w_grid - (ev_base + inner); m <- max(li)
    lp[j] <- m + log(max(sum(exp(li - m) * wt_trap), 1e-300))
  }
  exp(lp - max(lp)) / sum(exp(lp - max(lp)))
}

shares_of <- function(par15, label) {
  asc <- c(par15[1:5], 0); alpha <- c(par15[6:10], 0); sigma <- c(exp(par15[11:15]), 1)
  do.call(rbind, lapply(1:16, function(s) {
    prices <- c(as.numeric(price_matrix[s, c("GEN","GT","NAT","CHO","CAB")]), 0)
    data.frame(scenario = s, alternative = alt_labels,
               share = hev_probs_single(asc + alpha * prices, sigma), treatment = label)
  })))
}
```



```

shares <- bind_rows(shares_of(t4h$par_nh, "Nonhypothetical"),
                    shares_of(t4h$par_hyp, "Hypothetical")) |>
mutate(alternative = factor(alternative, levels = alt_labels),
       treatment   = factor(treatment, levels = c("Nonhypothetical", "Hypothetical")))

ggplot(shares, aes(scenario, share, colour = alternative)) +
  geom_line(linewidth = 0.8) + geom_point(size = 1.3) +
  facet_wrap(~treatment, ncol = 1) +
  scale_x_continuous(breaks = 1:16) +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1), limits = c(0, 1)) +
  scale_colour_brewer(palette = "Dark2", name = "Alternative") +
  labs(x = "Scenario", y = "Predicted choice probability",
       title = "Predicted market shares – HEV model (scenarios 1–16)") +
  theme_bw(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        strip.background = element_rect(fill = "grey90"))

```

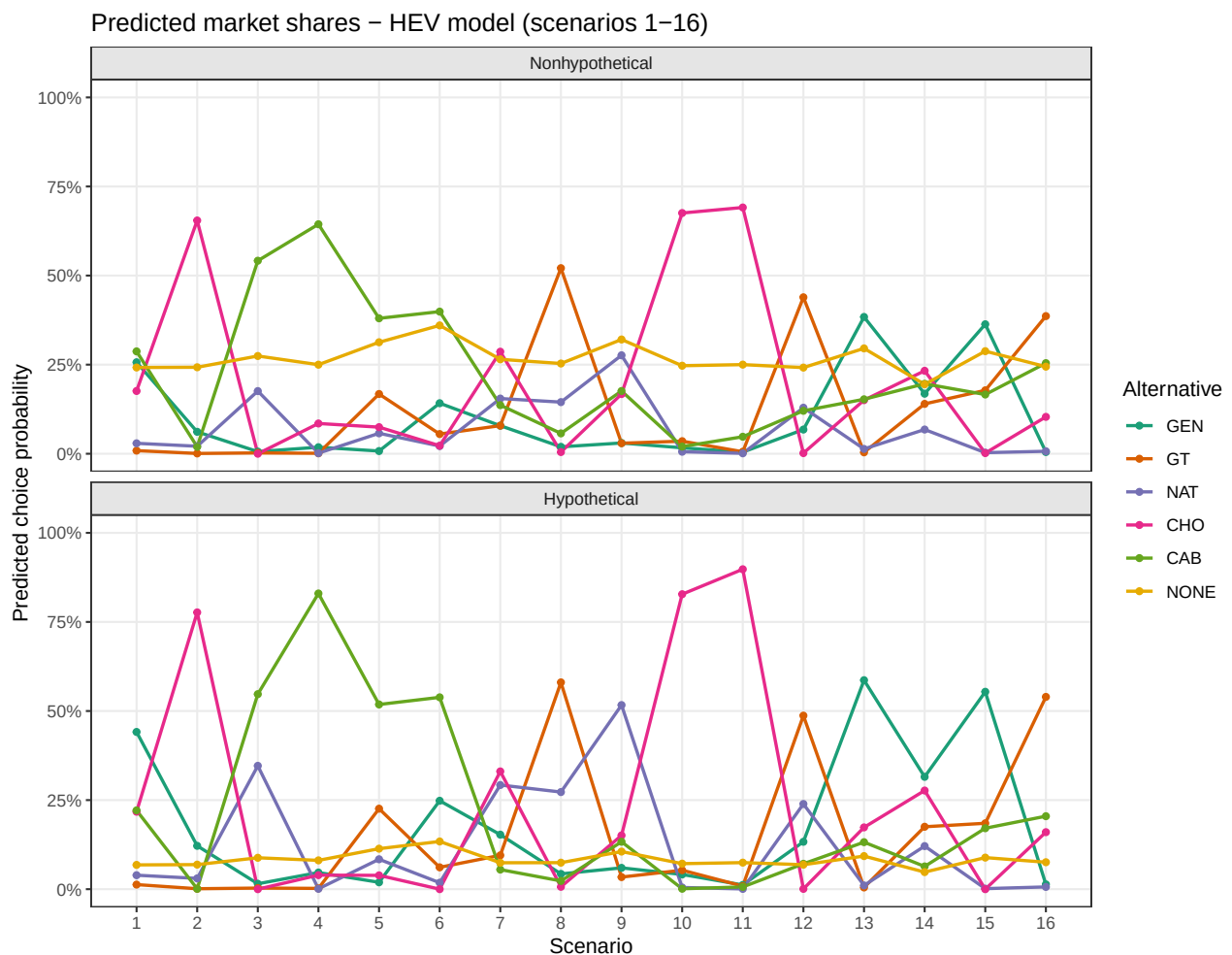


Figure 7.1: Figure 2. HEV-predicted market shares across scenarios, by treatment.

8 Replication experience and discrepancies

This chapter is the part the project asks for directly: what was hard, what I had to port from LIMDEP to R, and how I read the gaps between my numbers and the paper's.

8.1 Locating data and documentation

The data file (`harrison.xls`) and the questionnaire (`harrison.doc`) were both available, but neither was documented. The spreadsheet has a two-row header, label rows embedded in the middle, and summary rows at the bottom that look like data until you check them. The cleaning decision that mattered most, dropping subject 1060, is not obvious unless you go looking for rows with no choice answers, and that one step is what resolves the 68-versus-67 subject count.

The price design is another example. It lives only in the paper's appendix, not in the data file. It is the link between a stored choice code and the prices a respondent actually saw, so a single typo there would corrupt every estimate downstream. I keyed it in by hand and checked it against the appendix twice.

8.2 Translating LIMDEP → R

`mlogit` has no ready-made command for the scaled joint MNL or for HEV, so I wrote both as custom log-likelihoods and maximised them with `optim` (BFGS), using the log-sum-exp trick for stability and trapezoidal quadrature for the HEV Gumbel integral. In LIMDEP these are canned estimators. Writing them out by hand was useful for understanding, but it is also where small numerical differences creep in.

A second point is that the identification choices have to be made explicit in R. The base alternative (`NONE`), the fixed scale ($\mu_H = 1$), and the fixed HEV scale ($\sigma_{\text{NONE}} = 1$) are defaults in LIMDEP but have to be coded by hand here.

There was also one practical snag worth recording: `dfidx` 0.2.0 together with `vctrs` triggers a node-stack overflow, so I built the choice data with `mlogit.data()` instead.

8.3 Interpreting the discrepancies

Table 8.1: Documented discrepancies and interpretation.

Item	My replication	Paper	My reading
MNL/UL log-likelihood level	-1400 (nonhyp MNL)	-1261.4	Systematic level offset -> likely a price-utility specification difference (main open issue).
IIA test	101	31.3	Inflated by the LL-level offset; preference-equality test (differenced) matches (127 vs 131).
Chi-sq(20)	(nonhyp)	(p~0.051)	
MNL scale mu	0.804	0.90	Same sign & significance; optimiser/start-value sensitivity.
Scale-test df	9 (= (10+10)-11)	4	Bookkeeping of the joint restriction differs; conclusion (reject pooling) identical.
HEV sigma_CHO	~ 161 (NH)	moderate	Weak identification of Choice's error variance given the price design.
MNP	not estimated	estimated (apollo-style)	Needs GHK simulator (apollo); out of scope but documented.
RPL joint model	skipped (runtime)	estimated	Separate-model LLs used as the scale-test benchmark instead.

Taking stock, the qualitative side of the paper replicates cleanly. The two treatment groups are balanced on demographics; the real-payment group chooses “None” more often; IIA is rejected for the pooled and hypothetical data; and hypothetical WTP is above real WTP for every steak under every model. The numbers that do not match are limited to a few things: the log-likelihood level (and the IIA χ^2 that it inflates), the scale point estimate and the test degrees of freedom, and the one weakly identified HEV scale. None of these reflects a real disagreement with the paper; they come from optimiser defaults and from identification conventions that differ between LIMDEP and R.

i What I would chase next

The constant log-likelihood gap is the thread I would pull on with more time. The most likely cause is the price specification. I use five alternative-specific own-price coefficients α_j , but a single generic price coefficient, or a different way of scaling price, would change the likelihood I can reach while leaving the qualitative story untouched. Checking this against the paper's Table 3 specification and its footnote 13 would probably close most of the remaining gap.

9 Reproducibility

9.1 How to rebuild everything

The numbered scripts in `R/` produce the saved objects in `output/` that this book loads. To start over from the raw spreadsheet, run them in this order:

1. `R/01_data_prep.R` — read `harrison.xls`, clean, reshape; save `df_*.rds` and `price_matrix.rds`.
2. `R/02_table1_table2.R` — Tables 1 and 2.
3. `R/03_table3_mnl.R` — MNL estimates and the scale test → `table3_mnl.rds`.
4. `R/06_iaa_universal_logit.R` — IIA universal logit → `iaa_ul.rds`.
5. `R/04_table4_adv.R` — HEV and RPL → `table4_hev.rds`, `table4_rpl.rds`.
6. `R/05_table5_wtp.R` — WTP, Poe test, Figure 2 → `table5_wtp.rds`.

Then render the book:

```
quarto render      # builds the site into docs/
```

A couple of notes

The numbered scripts use `rstudioapi` to find the project folder, so run them from RStudio, or replace that line with `getwd()` to run them without RStudio.

The advanced models are slow (RPL with 500 draws, the HEV quadrature, the Poe bootstrap). The book does not re-run them; it loads `output/*.rds` and recomputes only the light summaries and Figure 2.

9.2 Session information

```
#> R version: R version 4.4.1 (2024-06-14)
#> Key packages:
#>  mlogit      1.1.3
#>  dfidx       0.2.0
#>  readxl      1.4.5
#>  dplyr       1.1.4
#>  tidyr       1.3.2
#>  ggplot2     3.5.1
#>  MASS        7.3.60.2
#>  Matrix      1.7.0
#>  knitr       1.49
```

References

- Lusk, J. L., & Schroeder, T. C. (2004). Are Choice Experiments Incentive Compatible? A Test with Quality Differentiated Beef Steaks. *American Journal of Agricultural Economics*, 86(2), 467–482.
- Poe, G. L., Giraud, K. L., & Loomis, J. B. (2005). Computational Methods for Measuring the Difference of Empirical Distributions. *American Journal of Agricultural Economics*, 87(2), 353–365.
- Swait, J., & Louviere, J. (1993). The Role of the Scale Parameter in the Estimation and Comparison of Multinomial Logit Models. *Journal of Marketing Research*, 30(3), 305–314.